

Gabriel Bridger

(+44) 07932276394 | gabriel.bridger@hotmail.com

<https://www.linkedin.com/in/gabrielbridger> | <https://gabriel-bridger.com>

Summary

Computer Science with AI student achieving First Class Honours. Experienced in building quantitative models for time series forecasting and statistical arbitrage using Python. Seeking a Quantitative Research internship to apply skills in machine learning and stochastic processes.

Education

University of Nottingham

BSc Computer Science with AI

June 2027

First Year Grade: First Class Honours

Relevant Courses:

Functional Programming in Haskell, Fundamentals of Artificial Intelligence, Developing Trading Algorithms, Mathematics for Computer Scientists - Advanced Linear Algebra and Applications.

Relevant Topics:

Time Series Analysis, Linear Algebra, Probability Theory, Functional & Object Oriented Languages, Type Theory

Burnley College | A Levels

AAB - Computer Science, Mathematics & Economics respectively

Work Experience & Projects

NEFS Quant | Quantitative Research Analyst

September 2025 - Present

- Selected as 1 of 14 from 200+ applicants to join the research analyst team, developing a strong conceptual understanding of quantitative finance.
- Competing in market-making games while developing skills in probabilistic thinking, game theory and quick mental maths.

Quantitative models for Time Series Forecasting | *Python, Shell, Scikit-Learn, XGBoost, NumPy, Pandas*

- Developed and compared multiple models to forecast stock returns, including an XGBoost implementation and more recently a Markov-Switching GARCH (MS-AR-GARCH) model built from scratch
- Engineered a parallelised Recursive Feature Elimination pipeline to optimise the input features
- Implemented a hybrid strategy using a Monte Carlo Simulation to produce Conditional Value at Risk based positions
- Architected a containerised backtesting system to enable rapid and efficient model evaluation

Principal Component Analysis for Statistical Arbitrage | *Python, NumPy, Pandas*

- Developed a mean-reversion trading strategy by applying Principal Component Analysis to a basket of 50 S&P 500 stocks
- Utilised the Marchenko-Pastur distribution to separate signal from noise and optimise the number of principal components used for the strategy
- Generated trading signals by regressing asset returns against factors and exploiting Z-scored residual spreads

Formula Student AI Path Planning Engineer | *C++, CMake, Git*

September 2025 - Present

- Engineered a track mapping algorithm in C++ using Delaunay Triangulation to generate a navigational mesh from cone coordinate data
- Developing a path optimisation solver to compute the minimum-curvature racing line

Skills

- **Languages:** Python, Haskell, C++, Git
- **Libraries/Frameworks:** TensorFlow/PyTorch, Scikit-Learn, Pandas, NumPy, Matplotlib, yfinance
- **Quantitative Concepts:** Time Series Analysis (LSTMs, RNNs, ANNs), Stochastic Processes, Principal Component Analysis, Computational Geometry